

2 Logic

2.1 Statements

Decide whether or not the following are statements. In the case of a statement, say if it is true or false, if possible.

2. Every even integer is a real number.

Statement, true.

4. Sets $\mathbb{Z}$ and $\mathbb{N}$.

Not a statement.

6. Some sets are finite.

Statement, true.

8. $\mathbb{N} \notin \mathcal{P}(\mathbb{N})$

Statement, false.

10. $(\mathbb{R} \times \mathbb{N}) \cap (\mathbb{N} \times \mathbb{R}) = \mathbb{N} \times \mathbb{N}$.

Statement, true.

12. If the integer x is a multiple of 7, then it is divisible by 7.

Statement, true.

14. Call me Ishmael.

Not a statement.

2.2 And, Or, Not

Express each statement or open sentence in one of the forms $P \wedge Q$, $P \vee Q$, or $\neg P$. Be sure to also state exactly what statements P and Q stand for.

2. The matrix A is not invertible.

$\neg P$, where P : Matrix A is invertible.

4. $x < y$.

$\neg P$, where P : $x \geq y$.

6. There is a quiz scheduled for Wednesday or Friday.

$P \vee Q$, where P : A quiz is scheduled for Wednesday, and Q : A quiz is scheduled for Friday.

8. At least one of the numbers x and y equals 0.

$P \vee Q$, where P : $x = 0$, and Q : $y = 0$.

10. $x \in A \cup B$.

$P \vee Q$, where P : $x \in A$, and Q : $x \in B$.

12. Happy families are all alike, but each unhappy family is unhappy in its own way. (*Leo Tolstoy, Anna Karenina*)

$P \wedge Q$, where P : Happy families are all alike, and Q : Each unhappy family is unhappy in its own way.

14. A man should look for what is, and not for what he thinks should be. (*Albert Einstein*)

$P \wedge \neg Q$, where P : Man should look for what is, and Q : Man should look for what he thinks should be.

2.3 Conditional Statements

Without changing their meanings, convert each of the following sentences into a sentence having the form “*If P , then Q* ”.

2. For a function to be continuous, it is sufficient that it is differentiable.

If a function is differentiable, then the function is continuous.

4. A function is rational if it is a polynomial.

If a function is a polynomial, then the function is rational.

6. Whenever a surface has only one side, it is non-orientable.

If a surface has only one side, then the surface is non-orientable.

8. A geometric series with ratio r converges if $|r| < 1$.

If a geometric series has a ratio with an absolute value less than 1, then the geometric series converges.

10. The discriminant is negative only if the quadratic equation has no real solutions.

If a quadratic equation has a discriminant that is negative, then the quadratic equation has no real solutions.

12. People will generally accept facts as truth only if the facts agree with what they already believe. (*Andy Rooney*)

If people accept facts as truth, then the facts agree with what they already believe.

2.4 Biconditional Statements

Without changing their meanings, convert each of the following sentences into a sentence having the form “ *P if and only if Q* ”.

2. If a function has a constant derivative then it is linear, and conversely.

A function is linear if and only if the function has a constant derivative.

4. If $a \in \mathbb{Q}$ then $5a \in \mathbb{Q}$, and if $5a \in \mathbb{Q}$ then $a \in \mathbb{Q}$.

$5a \in \mathbb{Q}$ if and only if $a \in \mathbb{Q}$

2.5 Truth Tables for Statements

Write a truth table for the logical statements in problems 1-9:

2. $(Q \vee R) \iff (R \wedge Q)$

Q	R	$(Q \vee R)$	$(R \wedge Q)$	$(Q \vee R) \iff (R \wedge Q)$
T	T	T	T	T
T	F	T	F	F
F	T	T	F	F
F	F	F	F	T

4. $\neg(P \vee Q) \vee (\neg P)$

P	Q	$(P \vee Q)$	$\neg(P \vee Q)$	$\neg P$	$\neg(P \vee Q) \vee (\neg P)$
T	T	T	F	F	F
T	F	T	F	F	F
F	T	T	F	T	T
F	F	F	T	T	T

6. $(P \wedge \neg P) \wedge Q$

P	Q	$(P \wedge \neg P)$	$(P \wedge \neg P) \wedge Q$
T	T	F	F
T	F	F	F
F	T	F	F
F	F	F	F

8. $P \vee (Q \wedge \neg R)$

P	Q	R	$(Q \wedge \neg R)$	$P \vee (Q \wedge \neg R)$
T	T	T	F	T
T	T	F	T	T
T	F	T	F	T
T	F	F	F	T
F	T	T	F	F
F	T	F	T	T
F	F	T	F	F
F	F	F	F	F

2.6 Logical Equivalence

A. Use truth tables to show that the following statements are logically equivalent.

2. $P \vee (Q \wedge R) = (P \vee Q) \wedge (P \vee R)$

P	Q	R	$(Q \wedge R)$	$P \vee (Q \wedge R)$	$(P \vee Q)$	$(P \vee R)$	$(P \vee Q) \wedge (P \vee R)$
T	T	T	T	T	T	T	T
T	T	F	F	T	T	T	T
T	F	T	F	T	T	T	T
T	F	F	F	T	T	T	T
F	T	T	T	T	T	T	T
F	T	F	F	F	T	F	F
F	F	T	F	F	F	T	F
F	F	F	F	F	F	F	F

4. $\neg(P \vee Q) = (\neg P) \wedge (\neg Q)$

P	Q	$(P \vee Q)$	$\neg(P \vee Q)$	$\neg P$	$\neg Q$	$(\neg P) \wedge (\neg Q)$
T	T	T	F	F	F	F
T	F	T	F	F	T	F
F	T	T	F	T	F	F
F	F	F	T	T	T	T

6. $\neg(P \wedge Q \wedge R) = (\neg P) \vee (\neg Q) \vee (\neg R)$

P	Q	R	$(P \wedge Q \wedge R)$	$\neg(P \wedge Q \wedge R)$	$\neg P$	$\neg Q$	$\neg R$	$(\neg P) \vee (\neg Q) \vee (\neg R)$
T	T	T	T	F	F	F	F	F
T	T	F	F	T	F	F	T	T
T	F	T	F	T	F	T	F	T
T	F	F	F	T	F	T	T	T
F	T	T	F	T	T	F	F	T
F	T	F	F	T	T	F	T	T
F	F	T	F	T	T	T	F	T
F	F	F	F	T	T	T	T	T

8. $\neg P \iff Q = (P \implies \neg Q) \wedge (\neg Q \implies P)$

P	Q	$\neg P$	$\neg Q$	$\neg P \iff Q$	$(P \implies \neg Q)$	$(\neg Q \implies P)$	$(P \implies \neg Q) \wedge (\neg Q \implies P)$
T	T	F	F	F	F	T	F
T	F	F	T	T	T	T	T
F	T	T	F	T	T	T	T
F	F	T	T	F	T	F	F

B. Decide whether or not the following pairs of statements are logically equivalent.

10. $(P \implies Q) \vee R$ and $\neg((P \wedge \neg Q) \wedge \neg R)$

P	Q	R	$(P \implies Q)$	$(P \implies Q) \vee R$	$\neg Q$	$(P \wedge \neg Q)$	$\neg R$	$(P \wedge \neg Q) \wedge \neg R$	$\neg((P \wedge \neg Q) \wedge \neg R)$
T	T	T	T	T	F	F	F	F	T
T	T	F	T	T	F	F	T	F	T
T	F	T	F	T	T	T	F	F	T
T	F	F	F	F	T	T	T	T	F
F	T	T	T	T	F	F	F	F	T
F	T	F	T	T	F	F	T	F	T
F	F	T	T	T	T	F	F	F	T
F	F	F	T	T	T	F	T	F	T

The statements are logically equivalent.

12. $\neg(P \implies Q)$ and $P \wedge \neg Q$

P	Q	$(P \implies Q)$	$\neg(P \implies Q)$	$\neg Q$	$P \wedge \neg Q$
T	T	T	F	F	F
T	F	F	T	T	T
F	T	T	F	F	F
F	F	T	F	T	F

The statements are logically equivalent.

14. $P \wedge (Q \vee \neg Q)$ and $(\neg P) \implies (Q \wedge \neg Q)$

P	Q	$\neg Q$	$(Q \vee \neg Q)$	$P \wedge (Q \vee \neg Q)$	$\neg P$	$(Q \wedge \neg Q)$	$(\neg P) \implies (Q \wedge \neg Q)$
T	T	F	T	T	F	F	T
T	F	T	T	T	F	F	T
F	T	F	T	F	T	F	F
F	F	T	T	F	T	F	F

The statements are logically equivalent.

2.7 Quantifiers

Write the following as English sentences, Say whether they are true or false.

2. $\forall x \in \mathbb{R}, \exists n \in \mathbb{N}, x^n \geq 0$

For every real number x , there is a natural number n for which $x^n \geq 0$.
The statement is true, consider $n = 2$.

4. $\forall X \in \mathcal{P}(\mathbb{N}), X \subseteq \mathbb{R}$

Every subset of the natural numbers is also a subset of the real numbers.
The statement is true, since $\mathbb{N} \subseteq \mathbb{R}$.

6. $\exists n \in \mathbb{N}, \forall X \in \mathcal{P}(\mathbb{N}), |X| < n$

There exists a natural number n for which $|X| < n$ for all subsets X of the natural numbers.
This statement is false, since $\mathbb{N} \in \mathcal{P}(\mathbb{N})$ and $\mathbb{N}$ is infinite.

8. $\forall n \in \mathbb{Z}, \exists X \subseteq \mathbb{N}, |X| = n$

For every integer n , there exists a subset X of the natural numbers for which $|X| = n$.
This statement is false, consider $n = -1$.

10. $\exists m \in \mathbb{Z}, \forall n \in \mathbb{Z}, m = n + 5$

There exists an integer m such that for every integer n , $m = n + 5$.
This statement is false, consider $n = 0$ and $n = 1$.